Case Study

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Case-Study

Module - I :-

1) Importance of SCM in Online Banking System:-

Software Configuration Management is a Systematic approach used to manage, control and track changes in Software Systems In a large scale Online Banking System, SCM is exteremely important due to the System's complexity, Continuous updates, and Security critical nature.

Firstly, Online banking applications are frequently updated with new features such as digital payments, loan processing, and fraud detection. without SCM, these updates can lead to Confusion, version conflicts, and loss of important code. SCM ensures that all changes are properly recorded and integrated in a structured manner.

Secondly, banking Systems are Safety critical Systems. Any failure in modules like fraud transfer of funds or interest calculation can result in financial loss and damage to customer trust. SCM helps maintain stable baselines, which acts as approved versions of the System. In case of failure the System can roll back to a previous stable version.

Another important ascept is security and compliance. Banking Systems must follow strict regulations. SCM ensures that only authorized changes are made, and every modification is documented. This helps in maintaining security and passing audits.

SCM also improves team coordination. Multiple teams work on different modules simultaneously. SCM tools help manage their work efficiently, avoiding conflicts and duplication.

Finally, SCM provides traceability, meaning every change debugging and maintenance.

In conclusion, SCM ensures reliability, consistency, security and efficient management, making it essential for an Online Banking System.

2) Versioning, Baselines and change Control

In large Systems like online banking, multiple teams often work on the same module, such as the funds - transfer module. without proper control, this can lead to integration issues and System instability.

Versioning is the process of maintaining different versions of a software component. Each update is stored as a new version. This allows developers to work independently without overwriting each other's work. If error occur, previous versions can be restored.

Baselines are approved versions of software that acts as a reference point. Once a baseline is established, it cannot be modified without proper approval. This ensures that all teams are working on a consistent and stable version of the System.

Change Control ensures that any modification to the System is reviewed and approved before implementation. This prevents unauthorized changes and reduces the risk of errors.

Together, these concepts help avoid integration issues by:

- Preventing conflicts between different versions
- Ensuring compatibility between modules
- Maintaining system stability.

For example, if two teams modify the same function, versioning tracks both changes, baselines provide a stable reference, and change control decides which change should be accepted.

Thus versioning, baselines and change control are essential for managing baselines parallel development effectively.

Module-2

3) Identifying Configuration Items (CIs)

Configuration Items (CIs) are the components of a System that need to be controlled and managed under SCM. In an Online Banking System, Identifying and classifying CIs is essential for maintaining Consistency and control.

The first step is to identify all artifacts involved in the project. These include not only source code but also documentation, testcases, and database systems.

CIs can be classified into the following categories.

1) Software CIs : Source code, APIs, backend logic.
2) Documentation CIs: Design documents, API documentation, user manuals.
3) UI CIs: Screens, webpages, layouts.
4) Test CIs: Test cases, scripts, Test reports.
5) Database CIs: Tables, schemas, stored procedures.

Each CI is assigned with a unique Identifier, version number, and owner. This helps in tracking changes and maintaining accountability.

Proper identification of CIs ensures:-

1) Better Control over System Components
2) Easy tracking of changes
3) Improved coordination among teams.

4) Change Control Board (CCB) process

The change control Board (CCB) is responsible for managing and approving changes in the system. When a customer reports an interest-calculation error, the CCB follows a structured process.

First, a change Request (CR) is submitted, which includes details such as the issue description, priority, and impact. Next, the CCB performs an evaluation of the request. It analyzes:

- → The severity of the issue
- → Risk involved
- → Cost and effort required
- → Impact on other modules

Based on this analysis, the CCB decides whether to approve or reject the request. If approved, the change is assigned to the development team.

The developers then implement the fix, ensuring that it does not affect other functionalities. After implementation, the fix is tested thoroughly.

Finally, the change is tracked and documentation in the SCM system using Status Accounting.

→ Reduction of errors, CCB ensures that all changes handled

Module-3

5) Role of Status Accounting:-

Status Accounting is an SCM activity that records and reports the status of configuration items and changes. It helps maintain transparency and control over the System.

In the given Scenario, a mismatch is found between the Production version and the baseline. Status accounting helps detect this by maintaining detailed records of

- versions of each CI
- Approved baselines
- Change history

By comparing the production System with recorded baseline, discrepancies can be identified easily. This helps detect unauthorized or accidental changes.

Status Accounting also provides:

- Traceability of changes
- Real time System status
- Supporting for audits and debugging.

Mismatch does not support status Accounting report.

Thus, it ensures consistency, accountability and better System control.

6) FCA and PCA:-

Functional Configuration Audit and physical Configuration Audit are two important SCM activities used to ensure System quality, used to verify correctness, Completeness and consistency of a System before release

Functional Configuration Audit (FCA):-

FCA focuses on verifying whether the System satisfies all functional requirements specified in the Software Requirement Specification (SRS). It ensures that the System behaves as expected and all functionalities are completely implemented.

→ During FCA, the following activities are performed:-

- verification of all functional Requirements
- Execution of test cases and validation of results
- checking Systems performance against expected outputs
- Ensuring that all user requirements are fulfilled.

In Banking System FCA,

→ fund transfer operations are executed correctly.
→ Interest calculations are accurate.
→ login and authentication mechanisms work properly.
→ Transactions are processed securely.

Physical Configuration Audit (PCA)

PCA focuses on verifying whether the physical components of the System match the documentation. It ensures that all configuration items (CIs) such as code, documents and design specifications are complete, consistent and properly recorded.

→ During PCA the following are checked:-

- Source code matches the approved design documents.
- All modules are present in the final build.
- Documentation is complete and upto date.
- Version numbers and configuration details are correct.

→ In Banking System, PCA is ⇒

- The deployed banking application version matches the documented release version.
- API documentation reflects the actual implementation.
- Database schema matches the design specifications.

PCA ensures that the system is properly documented and reproducible, which is essential for maintenance and future updates.

Module 4

7) CMMI and ITIL

CMMI and ITIL are frameworks that help improve SCM processes.

CMMI focuses on improving process maturity. It provides structured practices that make SCM processes more consistent and reliable.

ITIL focuses on IT service management. It improves areas such as change management and release management.

In an Online Banking System:-

CMMI ensures process standardization

ITIL ensures efficient service delivery.

→ These two frameworks helps to prevent the failures in critical operations like transactions and data security

→ ITIL provides a well defined change management framework, while CMMI ensures the process is followed consistently. This ensures that all changes are properly evaluated, approved, and implemented without affecting system stability.

8) COBIT Control Objectives

COBIT (control objectives for information and Related Technologies) is a globally recognized framework used for IT goverance and management. It provides a set of best practices and control objectives to ensure that IT Systems, especially in financial organizations like banking Systems, are Secure, reliable and compliant regulations.

1) change Management Control:-

This Control ensures that all changes made to the System are properly reviewed, approved, and documented before implemented.

Prevents:-

unauthorized or accidental changes
System instability
introduction of Security vulnerabilities.

2) Configuration Management Control:-

This Control focuses on maintaining accurate and upto date records of all System components, known as CIs.

ensures:-

1) System Consistency
2) Easy tracking of changes
3) Better maintenance and debugging.

9) SCM plan (SCMP)

Software configuration management plan (SCMP) is a formal document that defines how configuration management activities will be carried out throughout the software development life cycle. In an Online Banking System, an SCMP is essential to ensure that all changes are properly controlled, tracked and implemented in a systematic manner.

Key elements

1) Roles and Responsibilities:-

SCM Manager, developers, QA team, CCB

2) Baselines:-

Defined at key stages like after testing

3) Audit Frequency:-

Regular audits

4) Documentation:-

Version logs, Change Requests, audit reports.

10) SCM Team Structure:-

An effective SCM Team structure is essential for managing Configuration activities effectively in a large scale project like an Online Banking System. A well organized team ensures smooth handling of version control, build processes and software releases.

Key roles in SCM Team:-

1) Configuration Manager:-
   - Responsible for planning and managing SCM activities
   - Defines policies, procedures and standards.

2) Build Engineer:-
   - Manages Build processes
   - Automates compilation and integration.

3) Release Manager:-
   - Handles deployment of software versions
   - Ensures that only approved and tested versions are released.

Required Skills

1) Knowledge of version Control Tools
2) Automation Skills
3) Problem-solving ability
4) Communication Skills
5) understanding of Software lifecycle.

Key Activities performed:-

1) Build Management:-

Automating build processes

Ensuring consistent and error free builds

2) Version Control:-

Managing different versions of code

Handling branching and merging.

3) Release Management:-

Planning and Scheduling releases

Ensuring stable and tested deployments.

Module 6

ii) SCM Tool Comparison:-

SCM Tools are essential for managing changes, managing versions and maintaining and ensuring smooth collaboration in software projects.

Two Tools are Git and Subversion (SVN):-

1) Git (Distributed Version Control system):-

Git is a distributed version control system where each developer has a complete copy of the repository on their local machine.

Key features:-

1) Distributed Nature
2) Fast performance
3) Branching and Merging
4) High Flexibility.

2) SVN (centralized Version Control System):-

SVN is a centralized version control system where all files are stored in a central repository.

Key features:-

1) centralized Repository
2) Simpler model
3) Maintains a single source of truth.

12) Role of SCM Tools:-

SCM Tools play a vital role in automating and managing software development activities. In an Online Banking System, tools such as git, Jenkins and Product data management (PDM) System helps ensure efficient and error free configuration management

1) Git (version Control Tool):-

Used to manage Source code versions and track changes.

Benefits

- easy rollback to previous versions
- Prevents code conflicts
- Improves team collaboration.

2) Jenkins (Build Automation Tool):-

Jenkins is a continuous Integration | continuous deployment CI|CD tool used to automate build and testing process

- Runs test cases to detect errors early.
- Integrates different modules.

3) PDM (product Data Management):-

PDM Systems manages all product related data and documentation.

- Improves data organisation
- Ensures accurate documentation